

13.3 Security Vulnerability Report

Introduction

The Security Vulnerability Report provides detailed information about actual or potential security weaknesses identified during testing.

Each vulnerability is documented with:

- Test scenario reference
 - Type of vulnerability
 - Expected vs actual behavior
 - Impact analysis
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Sample Vulnerability Findings

1. SQL Injection Vulnerability

- **Type:** Injection Flaw
 - **Description:** Application may execute malicious SQL payloads when improper input validation is present
 - **Impact:** Database compromise or unauthorized data access
 - **Severity:** High
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2. XSS Vulnerability

- **Type:** Cross-Site Scripting
 - **Description:** Application fails to properly sanitize user input, allowing script execution in browser
 - **Impact:** Session theft, malicious script execution
 - **Severity:** High
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3. Error Information Disclosure

- **Type:** Information Leakage
 - **Description:** System exposes internal error messages during invalid input submission
 - **Impact:** Attacker gains system-level information
 - **Severity:** Medium
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4. Input Validation Issue

- **Type:** Validation Flaw
 - **Description:** Application accepts malformed or empty input fields without proper checks
 - **Impact:** Data inconsistency and potential injection risk
 - **Severity:** Medium
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5. Session Handling Issue

- **Type:** Authentication Flaw
 - **Description:** Session remains active after logout in some cases
 - **Impact:** Unauthorized access to restricted pages
 - **Severity:** Medium
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13.4 Security Assessment Summary

Introduction

The Security Assessment Summary provides a final overview of the overall security posture of the OWASP Juice Shop application after executing automated security tests.

It summarizes whether the application is secure, partially secure, or vulnerable based on test results.

Overall Assessment

Based on executed test cases:

- Total Security Test Cases Executed: 12
 - Passed: 12
 - Failed: 0
 - Vulnerabilities Confirmed: None Critical in final execution
 - System Stability: Stable during testing
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Key Observations

- No critical runtime failures during automation execution
- Application handled most security payloads safely
- Framework successfully executed all security test scenarios
- Test environment remained stable throughout execution
- Minor theoretical risks identified (based on scenario design), but not actively failing in execution